



Multimodality as Bridge or Hurdle in Human–AI Communication

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- Humans often infer more than the literal content of an image; what does it communicate?
- Meaning may depend on context, shared knowledge, and abstraction; AI often only captures surface associations

Example:

- Literal description: *“two children in a park”*
- Underlying meaning: *friendship*

Question: How can we study this gap in communication systematically?



Figure: Literal scene, but also a possible abstract meaning.

- When does multimodality help communication?
- When does it make communication harder?
- What expectations do people have regarding AI capabilities in the context of multimodal communication?

Multimodality as bridge

- one modality compensates for another
- helps when a concept is hard to express in words alone
- can support abstraction and communication

Multimodality as hurdle

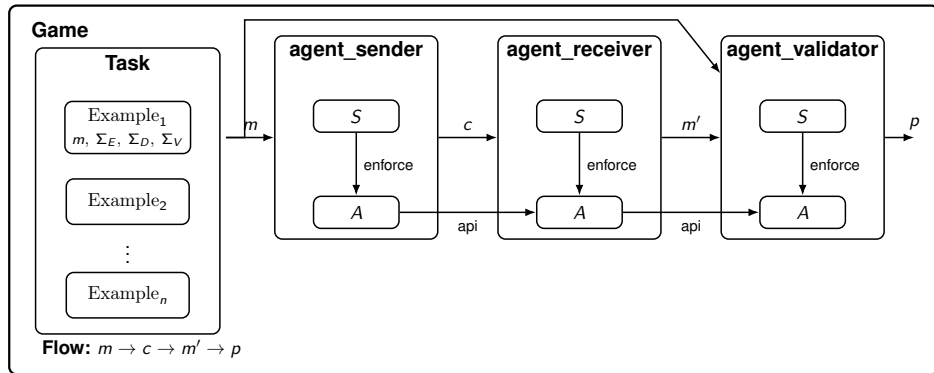
- additional meaning has to be inferred
- context and association matter
- sender and receiver may align differently

- Many **multimodal communication tasks** are gamelike and follow a similar framework
- They separate three roles:
 - **sender**: encodes a concept
 - **receiver**: decodes it
 - **verification**: checks success

Why useful here?

- this abstraction captures a large variety of games
- the “difficulty”, serves as bridge/hurdles





The framework separates the task definition, agent behavior, interface constraints, and round-level control flow.

Alphabets

Σ_E : encoder alphabet, Σ_D : decoder alphabet

En/Decode with emojis/ images rather than letters

Signatures define valid outputs

single choice: $x \in \Sigma$, multiple choice: $X \subseteq \Sigma$, word sequence: Σ^+

More restriction on choice, limits en/decoder

We make games hard, to see change in en/decoder behavior under different conditions

Expectations regarding AI capabilities



Via captioning task

- We instantiate the framework with a captioning game over **image pairs**
- For each pair, a human encoder selects one image as the target message m
- The encoder then produces a caption c that should help a receiver identify the target

Two conditions

- 1 **Standard captioning:** write a caption so that another human can identify the target image
- 2 **AI-aware captioning:** write a caption that remains understandable for humans, but should be harder for AI to decode

Assumption: Images have many hidden meaning; human might have expectations what other humans might decode but AI doesn't

- **30 images** organized into **15 visually related pairs**
- Three categories:
 - **Internet memes** (cultural knowledge, humor, implicit context)
 - **Stock photography** (literal and prototypical concepts)
 - **Abstract art** (weak object-level semantics, more interpretive abstraction)

Pairing strategy

- **Abstract art:** pairs from the **same artist**
- **Stock images:** photos from the **same search term**
- **Memes:** paired by **close CLIP embeddings**

Goal: create pairs with different types of similarity and ambiguity to test selective interpretability.

Live Demo!

	Human raters	AI raters	Human - AI
baseline	Fleiss' κ	Fleiss' κ	Cohen's κ
AI-aware	Fleiss' κ	Fleiss' κ	Cohen's κ
accuracy	Accuracy vs. ground truth	Accuracy vs. ground truth	

Expectation

- AI-aware condition should make decoding harder for both humans and AI
- Humans should remain more accurate and consistent than AI
- The gap may reflect how well humans anticipate AI's capabilities

Metric	Standard	AI-aware
Human Fleiss' κ	0.954	0.636
AI Fleiss' κ	0.591	0.679
Human–AI Cohen's κ	0.167	-0.024
Mean human accuracy	98.3%	85.0%
Mean AI accuracy	52.2%	43.3%
AI majority accuracy	53.3%	40.0%

- Harder captions hurt **both** humans and AI
- But they also increase **human–AI divergence**

The hurdle changes not only performance, but interpretation.

Humans

κ

0.954 $\rightarrow$ 0.636

- shorter captions
- less literal
- more ambiguous

AI

accuracy

52.2% $\rightarrow$ 43.3%

- fewer object cues
- more abstract wording
- harder to anchor

The captions become less literal — and harder to decode.

AI self-agreement

$$0.591 \xrightarrow{\kappa} 0.679$$

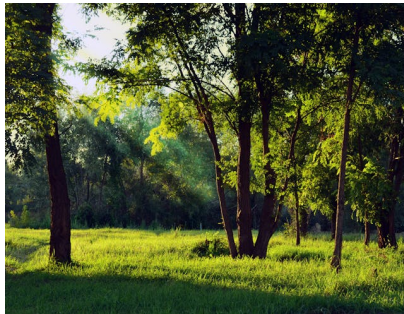
- not random failure
- more similar mistakes

Human–AI agreement

$$0.167 \xrightarrow{\kappa} -0.024$$

- different cues
- different interpretations

AI becomes less human-aligned, not simply more random.



Standard

[Forest and grass with sunshine, some shadow in the foreground and in the background more dense forest]

longer, literal, descriptive

AI-aware

[Fairy Tale]

shorter, indirect, evocative

Humans seem to shift from scene description to feeling, metaphor, or association.

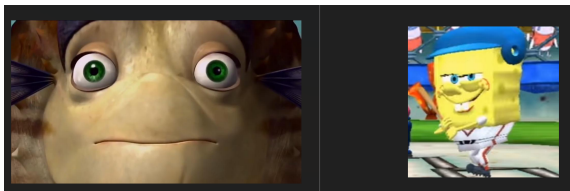
- familiar concepts can dominate
- AI may choose what it already “knows”
- caption matters less than prior salience

Example:

SpongeBob

Reading:

stable, but biased decoding



- Small dataset and participant number
- Ground truth serves as both reference label and part of the human evaluation setup
- Ai ratings are repeated runs from the same predated model
- Agreement metrics such as κ are sensitive to label prevalence

- Framework that captures multimodal games
- AI-aware captions made the task harder for everyone
- Humans still outperformed AI
- Human–AI agreement became even weaker
- Humans used more indirect captions:
 - less object-based
 - more mood, metaphor, association

**The hurdle reduced success,
and also pushed humans and AI toward different interpretations.**